Part A – Theory Answers

Q1. Explain Mean, Median and Mode in the context of customer income.

Mean, Median and Mode are measures of central tendency used to describe the central value of data such as customer income.

Mean

Mean is the **average income** of all customers. It is calculated by adding all income values and dividing by the number of customers.

Formula:

$$Mean = \frac{\sum Income}{Number\ of\ Customers}$$

Example:

Customer incomes: ₹30,000, ₹40,000, ₹50,000

$$Mean = \frac{30000 + 40000 + 50000}{3} = ₹40,000$$

Application:

Mean helps banks understand the **average earning level of customers**.

Median

Median is the **middle-income value** when all incomes are arranged in ascending order.

Example:

₹25,000, ₹30,000, ₹40,000, ₹50,000, ₹70,000

Median = ₹40,000

If values are even:

₹20,000, ₹30,000, ₹40,000, ₹50,000

Median:

$$Median = \frac{30000 + 40000}{2} = ₹35,000$$

Application:

Median is useful when extreme income values exist.

Mode

Mode is the **most frequently occurring income value**.

Example:

₹20,000, ₹30,000, ₹30,000, ₹40,000

Mode = ₹30,000

Application:

Mode shows the **most common income group**.

Q2. Differentiate between Standard Deviation and Variance using loan amounts.

Variance and Standard Deviation measure how much loan amounts vary from the average loan.

Variance

Variance measures the **spread of loan amounts** by calculating squared differences from the mean.

Formula:

$$Variance = \frac{\sum (x - mean)^2}{n}$$

Example:

Loan Amounts: ₹10,000, ₹20,000, ₹30,000

Mean = ₹20,000

Variance measures how far each value is from ₹20,000.

Standard Deviation
Standard deviation is the **square root of variance** and shows spread in the **same unit as loan amount**.
Formula:
$$ Standard\ Deviation = \sqrt{Variance} $$
Example:
If Variance = 40000
$$ Std\ Dev = \sqrt{40000} = 200 $$

Difference

Variance	Standard Deviation
Square of differences	Square root of variance
Unit is squared	Same unit as data
Hard to interpret	Easy to interpret

Q3. What is a Random Variable? Give one example from the dataset.
A **Random Variable** is a variable whose value depends on the outcome of a random event.
Random variables can take different values depending on chance.
Example from Dataset
Default Status
It has two possible values:
Default = Yes
No Default = No
We can represent:
Yes = 1
No = 0
Since default depends on customer behaviour, it is considered a **random variable**.
Another example:
Loan Amount (continuous random variable).

Q4. Explain Conditional Probability in terms of loan defaults.
Conditional Probability is the probability of an event occurring **given that another event has already occurred**.
Formula
$$ P(A|B) = \frac{P(A \cap B)}{P(B)} $$
Where:
A = Event A
B = Event B

Example
Probability of default **given credit score < 600**
$$ P(Default \mid Credit\ Score < 600) $$
Suppose:
Customers with Credit Score < 600 = 100
Defaults among them = 40
$$ P(Default \mid Credit\ Score<600) = \frac{40}{100} = 0.4 $$

Application:
Banks use this to **predict loan risk**.

Q5. Define Bayes Theorem and mention how banks can apply it.
Bayes Theorem is used to update the probability of an event based on new information.

Formula
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$
Where:
P(A|B) = Probability of A given B
P(B|A) = Probability of B given A
P(A) = Prior probability
P(B) = Probability of B

Example
A bank calculates:
Probability of default **given low credit score**.
If:
P(Default) = 0.2
P(Low Score | Default) = 0.7
Bayes theorem helps calculate updated probability.

Application in Banking
Banks use Bayes theorem for:
- Credit risk prediction
- Loan approval
- Fraud detection

Q6. Differentiate between Empirical Probability and Theoretical Probability with examples.
Empirical Probability
Empirical probability is calculated using **real observed data**.
Formula:
$$Empirical\ Probability = \frac{Observed\ Events}{Total\ Events}$$
Example:
If 200 customers defaulted out of 1000:
$$P(Default) = \frac{200}{1000} = 0.2$$
This means **20% default rate**.

Theoretical Probability
Theoretical probability is calculated using **mathematical assumptions**.
Example:
Probability of getting Head in coin:
$$P(Head) = \frac{1}{2}$$

Difference

Empirical	Theoretical
Based on real data	Based on theory
Uses past records	Uses assumptions
Used in business analysis	Used in probability models

Q7. What is a Poisson Distribution? Give a business example.

A **Poisson Distribution** measures the number of times an event occurs in a fixed time interval.
It is used when:

- Events occur randomly
- Average rate is known

Formula

$$P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}$$

Where:
λ = average number of events
k = number of events

Business Example

Suppose:
A bank receives **5 loan applications per hour**.
Poisson distribution helps find:
Probability of receiving **3 applications in one hour**.
Application:
Used in:

- Queue management
- Customer arrival prediction
- Loan request analysis

Q8. Write a short note on Eigenvalues and Eigenvectors in data analysis.

Eigenvalues and Eigenvectors are important concepts in **linear algebra** used in data analysis.
Eigenvectors represent **directions of maximum variance**, while eigenvalues represent **magnitude of variance**.

Example in Data Analysis

In large datasets with many variables like:

- Income
- Loan Amount
- Credit Score

We use **Principal Component Analysis (PCA)**.
PCA uses eigenvalues and eigenvectors to:

- Reduce number of variables
- Identify important patterns
- Improve model performance

Thus, eigenvalues and eigenvectors help simplify complex datasets.